

Dr. Kousik Kumar Dutta

Computer Science Researcher | Open to Collaboration

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Profile

I am a computer science researcher working on preference-aware routing, time-dependent road networks, and scalable graph algorithms. I completed my PhD defence in Computer Science and Engineering at IIT Ropar in September 2026, after doctoral work on constrained maximization of preferences on time-dependent road networks. My research combines algorithms and systems, with an evolving agenda in database systems, spatial and network data management, scalable computing, and graph query processing. I am open to collaboration on graph algorithms, data systems, transportation networks, and related teaching or research initiatives.

Research directions

Constrained and Preference-Aware Path Query Processing

Efficient approaches for constrained and preference-aware path queries in large transportation networks.

Scalable Graph Representations and Path-Closure Processing

Graph representations and query-processing techniques for connectivity and path relationships in large networks.

GPU-Accelerated Graph Analytics and Query Processing

Parallel and hardware-accelerated approaches for large-scale graph workloads.

Education

Doctoral study in Computer Science and Engineering - Indian Institute of Technology Ropar (2021-2026)

Dissertation: Constrained Maximization of Preferences on Time-Dependent Road Networks. Defence completed 16 September 2026. Advisors: Dr. Venkata M. V. Gunturi and Dr. T. V. Kalyan. CGPA 7.92/10.00.

MTech, Computer Science and Engineering - Indian Institute of Technology Ropar (2019-2021)

Thesis: A Multi-Threading Algorithm for Constrained Path Optimization on Road Networks. CGPA 8.39/10.00.

BTech, Computer Science and Engineering - Kalyani Government Engineering College (2014-2018). CGPA 7.45/10.00.

Teaching and service

Teaching assistantships at IIT Ropar: CS301 Databases, CS305 Software Engineering, and CS509 PG Software Lab (2020-2026). Reviewer for GeoInformatica since 2023. Technical Committee Head for SYNTACS 2026. IEEE TCDE Student Grant recipient for MDM 2026. Institute fellowship, 2021-2026; MHRD GATE Scholarship, 2019-2021; GATE 2019 AIR 644.

Selected publications

2026 A Parallelizable Algorithm for Constrained Maximization of Preferences in Large Time-Dependent Graphs
GeoInformatica 30, article 19. DOI: 10.1007/s10707-026-00577-z.

2026 A User-Configurable Navigation System for Wideness and Turn-Aware Routing
MDM, pp. 339-342. DOI: 10.1109/MDM71479.2026.00050.

2025/2026 Interval Based Constrained Path Optimization in Time-Dependent Road Networks
WISE 2025 proceedings, LNCS 16368, pp. 165-176. DOI: 10.1007/978-981-95-7251-9_12.

2025 Constrained Path Optimization on Time-Dependent Road Networks
WISE 2024 Posters and Demos Track, LNCS 15463, pp. 274-282. DOI: 10.1007/978-981-96-1483-7_24.

2022 A Multi-Threading Algorithm for Constrained Path Optimization Problem on Road Networks
WISE 2022, LNCS 13724, pp. 110-118. DOI: 10.1007/978-3-031-20891-1_9.

2021 NEAT Activity Detection Using Smartwatch at Low Sampling Frequency
UIC track, pp. 25-32. DOI: 10.1109/SWC50871.2021.00014.

2021 A Fairness Conscious Cache Replacement Policy for Last Level Cache
DATE, pp. 695-700. DOI: 10.23919/DATE51398.2021.9474096.

2023 CAMOUFLAGE: An Efficient Mechanism to Hide Congestion in NVM LLC
HiPC 2023 Student Research Symposium extended abstract.

Current manuscripts

A Bi-Directional Search for Interval Constrained Path Optimization in Large Time-Dependent Road Networks
Knowledge and Information Systems - major revision submitted.

A Decomposition-Based Heuristic for Selective Pickup-Delivery Routing with Non-Additive Loading-Unloading Costs
Annals of Operations Research - under review.

Profiles

Google Scholar: scholar.google.com/citations?hl=en&user=cVei1KAAAAAJ | DBLP: dblp.org/pid/297/4783.html | ORCID: orcid.org/0000-0003-0779-1354 | GitHub: github.com/kousik-kr